Autodesk®
Scaleform®

Scaleform 4.2

Scaleform 4.2

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GAMEWARE

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Autodesk® Scaleform® 4.2

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Scaleform 4.0 (Advance) Scaleform SDK Scaleform 4.0 API Scaleform

2

API Scaleform 3.x Scaleform 4.0

1. Renderer2D
2. GFxLoader
3.
4. MovieDisplayHandle Renderer2D::Display
5. GFx::Movie
6. API GRenderer Scaleform 3.X Render::HAL

2.1

Scaleform Render::Renderer2D Render::HAL Renderer2D Display Render::HAL Renderer2D “ ” Render::D3D9::HAL Render::PS3::HAL

```
D3DPRESENT_PARAMETERS   PresentParams;
IDirect3DDevice9*       pDevice = ...;
ThreadId                renderThreadId = Scaleform::GetCurrentThreadId();
Render::ThreadCommandQueue *commandQueue = ...;
Ptr<Render::D3D9::HAL>  pHal;
Ptr<Render::Renderer2D> pRenderer;

pHal = *SF_NEW Render::D3D9::HAL(commandQueue);
```

```
    if (!pHal-> InitHAL(Render::D3D9::HALInitParams(pDevice, PresentParams, 0,
                                                     renderThreadId)))
    {
       return false;
    }
    pRenderer = *SF_NEW Render::Renderer2D(pHal);
```

HALInitParams class pDevice
PresentParams renderThreadId
ThreadCommandQueue

Renderer2D HAL

2.2

Scaleform 4.0 Renderer2D
Scaleform 3.x GFxLoader InitHAL
Render::HAL ShutdownHAL

Render::HAL HAL

```
    MeshCacheParams mcp;
    mcp.MemReserve          = 1024 * 1024 * 3;
    mcp.MemLimit            = 1024 * 1024 * 12;
    mcp.MemGranularity      = 1024 * 1024 * 3;
    mcp.LRUTailSize         = 1024 * 1024 * 4;
    mcp.StagingBufferSize   = 1024 * 1024 * 2;

    pRenderer->GetMeshCacheConfig()->SetParams(mcp);
```

MemReserve MemLimit
MeshCache MemGranularity
LRU LRUTailSize MeshCache StagingBufferSize
64K PC

```
GlyphCacheParams gcp;
  gcp.TextureWidth  = 1024;
  gcp.TextureHeight = 1024;
  gcp.NumTextures   = 1;
  gcp.MaxSlotHeight = 48;

  pRenderer->GetGlyphCacheConfig()->SetParams(gcp);
```

Scaleform 3.3 TextureWidth TextureHeight NumTextures
MaxSlotHeight

InitHAL InitHAL SetParams

2.3

Scaleform 4.0 GFx::ImageCreator Render::TextureManager
Scaleform ImageCreator GFx::Loader

```
GFx::Loader loader;
...

SF::Ptr<GFx::ImageCreator> pimageCreator =
    *new GFx::ImageCreator(pRenderThread->GetTextureManager());
loader.SetImageCreator(pimageCreator);
```

ImageCreator TextureManager
TextureManager
Render::HAL::GetTextureManager()
GetTextureManager

Render::HAL TextureManager CreateTexture

- D3DCREATE_MULTITHREADED D3D9
- ThreadCommandQueue Render::HAL

D3D
ThreadCommandQueue
Platform::RenderHALThread

2.4

/

Scaleform 4.0 APIs

GFx::MovieDisplayHandle GFx::Movie

Display

1. GFx::MovieDef::CreateInstance GFx::Movie GFx::Movie
2. Advance ActionScript Invoke/DirectAccess API Advance Advance Scaleform
3. MovieDisplayHandle

```
    //----------------------------------------------------------------------
// 1.

    Ptr<GFx::Movie>          pMovie = ...;
    GFx::MovieDisplayHandle  hMovieDisplay;

//
//
    pMovie->SetViewport(width, height, 0,0, width, height);
    hMovieDisplay = pMovie->GetDisplayHandle();

//                                     Scaleform Player
//
    pRenderThread->AddDisplayHandle(hMovieDisplay);


    //----------------------------------------------------------------------
// 2.

//
//    ExternalInterface           Advance
    float deltaT = ...;
    pMovie->HandleEvent(...);
    pMovie->Advance(deltaT);
```

```
//
//
    pRenderThread->WaitForOutstandingDrawFrame();
    pRenderThread->DrawFrame();


    //------------------------------------------------------------------------
// 3.      –         DrawFrame
    Ptr<Render::Renderer2D> pRenderer = ...;

    pRenderer->BeginFrame();
    bool hasData = hMovieDisplay.NextCapture(pRenderer->GetContextNotify());
    if (hasData)
        pRenderer->Display(hMovieDisplay);
    pRenderer->EndFrame();
```

- WaitForOutstandingFrame – CPU

  Advance
- MovieDisplay.NextCapture – “ ”Advance

  false

2.5

GFx::Movie Release

Render::ThreadCommandQueue Movie
ThreadCommandQueue

Movie::ShutdownRendering(false)
Movie::IsShutdownRenderingComplete
NextCapture false

3

Scaleform –

GFx::Movie::Advance ActionScript
Advance Advance
Scaleform Advance

Render::HAL Scaleform

-
-

Scaleform Player Scaleform

3.1

Advance
/

Scaleform
GFx::Movie::Advance GFx::Movie::Capture

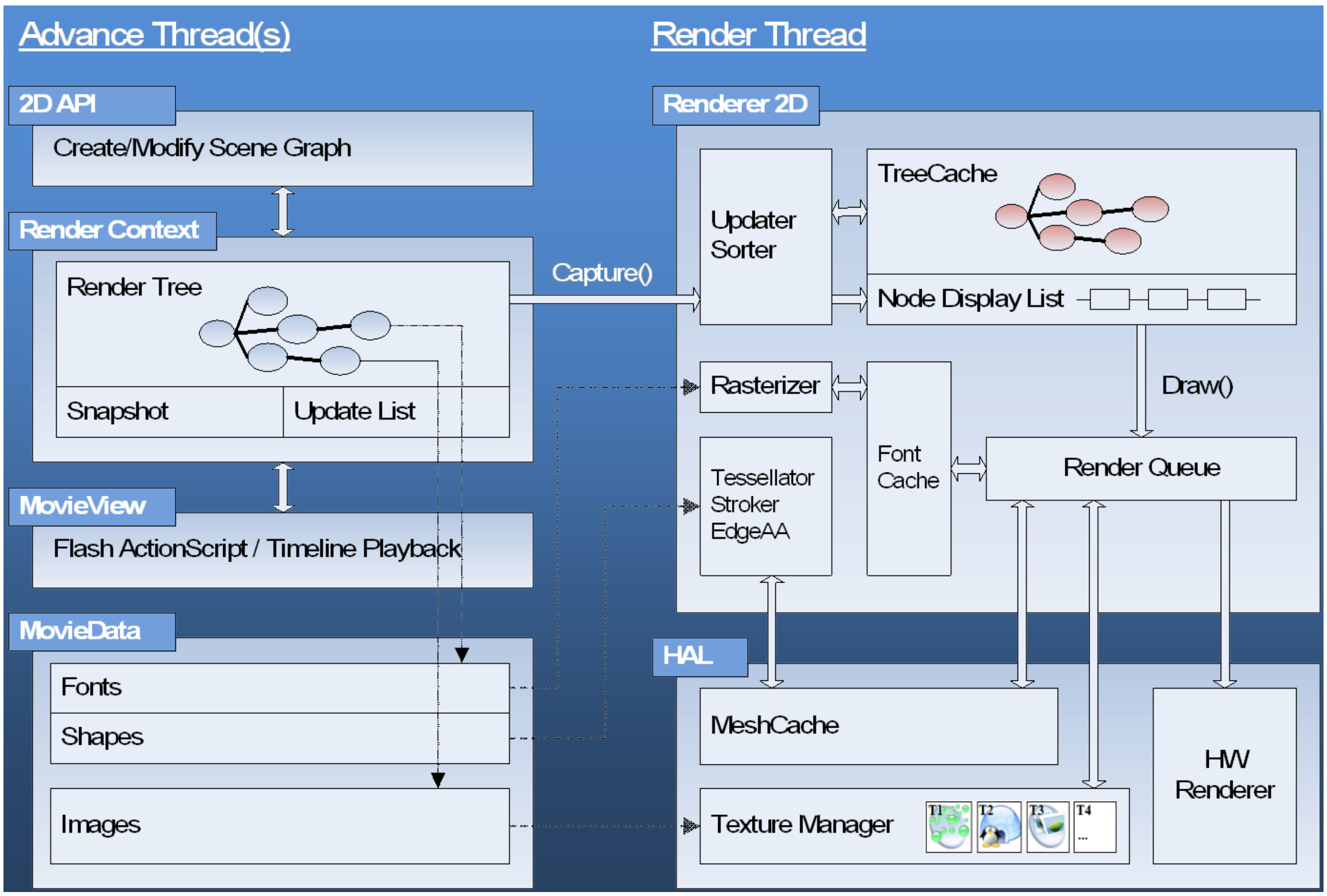
NextCapture

- Movie::Advance
- Capture NextCapture Capture Capture
- Capture Advance Invoke Direct Access API Movie

Scaleform 4.0 Scaleform 3.3 Display Movie::Invoke Capture Advance Advance Invoke

4

Scaleform 4.0



1

Movie MovieDef
Scaleform Render Context Render Tree
2D Scaleform Movie

Render::Renderer2D Render::HAL Renderer2D
Display Render::HAL
Render HAL

-
- Font Cache Font Cache
- Mesh Cache
- Render Queue Render Queue /
  CPU
- HAL API

ActionScript AI
/

5

Scaleform

5.1

ActionScript

AS2 “rendererString” “rendererFloat” AS3 “setRendererString”

“setRendererFloat” “AS2/AS3 ”

/ (Float)

HAL::PushUserData UserDataState::Data Render::HAL UserDataStack

“ ”

/ “ ” UserDataState::Data HAL::PopUserData

5.2

HAL

HAL

2D

AS3 SWF “m” AS3

```
import scaleform.gfx.*;
DisplayObjectEx.setRendererString(m, "Abc");
DisplayObjectEx.setRendererFloat(m, 17.1717);
```

4.1

Src/Render/ShaderData.xml

```
<ShaderFeature id="Position">
  <ShaderFeatureFlavor id="Position2d" hide="true"/>
  <ShaderFeatureFlavor id="Position3d"/>
  <ShaderFeatureFlavor id="Position2dOffset"/>
</ShaderFeature>
```

“Position2dOffset”

“Position2dOffset”

```
 <ShaderSource id="Position2dOffset" pipeline="Vertex">
    Position2d(
    attribute float4 pos    : POSITION,
    varying   float4 vpos   : POSITION,
    uniform   float4 mvp[2],
    uniform   float  poffset)
    {
    vpos = float4(0,0,0,1);
    vpos.x = dot(pos, mvp[0]) + poffset;
    vpos.y = dot(pos, mvp[1]) + poffset;
    }
  </ShaderSource>
```

“Position2d” – ‘poffset’ x y

D3D9_Shader.cpp ShaderInterface::SetStaticShader

```
//
    if (pHal->UserDataStack.GetSize() > 0 )
    {
        const UserDataState::Data* data = pHal->UserDataStack.Back();
        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
            data->RendererString.CompareNoCase("abc") == 0)
        {
//
            vshader = (VertexShaderDesc::ShaderType)((int)vshader +
VertexShaderDesc::VS_base_Position2dOffset);
            shader  = (FragShaderDesc::ShaderType)  ((int)shader +
FragShaderDesc::FS_base_Position2dOffset);
        }
    }
```

HAL UserDataStack “abc”
Position2dOffset
D3D9_ShaderDescs.h ShaderData.xml

‘poffset’
ShaderInterface::Finish

```
//
    if (pHal->UserDataStack.GetSize() > 0 )
    {
        const UserDataState::Data* data = pHal->UserDataStack.Back();
```

```
        if (data->Flags &
(UserDataState::Data::Data_Float|UserDataState::Data::Data_String) &&
            data->RendererString.CompareNoCase("abc") == 0)
        {
//    poffset
            for ( unsigned m = 0; m < Alg::Max<unsigned>(1, meshCount); ++m)
            {
                float poffset = data->RendererFloat / 256.0f;
                SetUniform(CurShaders, Uniform::SU_poffset, &poffset, 1, 0, m);
            }
        }
    }
```

'meshCount'
meshCount 1 HAL 'm'

5.3

HAL Scaleform

Scaleform
Scaleform / AS2
"disableBatching" AS3 "disableBatching" AS2/AS3

6 GFxShaderMaker

6.1

Scaleform 4.1 GFxShaderMaker Scaleform Render_D3D9 ShaderData.xml / D3D9

```
Src/Render/D3D9_ShaderDescs.h
Src/Render/D3D9_ShaderDescs.cpp
Src/Render/D3D9_ShaderBinary.cpp
```

6.2 API HAL

6.2.1 D3D9

D3D9 HAL ShaderModel 2.0 ShaderModel 3.0 HAL

GFxShaderMaker '-shadermodel'

```
GFxShaderMaker.exe –platform D3D1x –shadermodel SM30
```

ShaderModel 2.0 ShaderModel 2.0 InitHAL

6.2.2 D3D1x

D3D1x HAL

D3D11 D3D10.1

`FEATURE_LEVEL_10_0`

`FEATURE_LEVEL_9_3`

`FEATURE_LEVEL_9_1`

HAL

GFxShaderMaker '-

featurelevel'

```
GFxShaderMaker.exe –platform D3D1x –featurelevel
        FEATURE_LEVEL_10_0,FEATURE_LEVEL_9_1
```

FEATURE_LEVEL9_3 FEATURE_LEVEL_9_3

InitHAL FEATURE_LEVEL_9_1